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Course Teacher

NEURAL NETWORKS

1. INTRODUCTION TO NEURAL NETWORKS

A **Neural Network** is an important technology in **Artificial Intelligence (AI)** and **Machine Learning (ML)**. It is inspired by the way the human brain processes information. A neural network consists of interconnected artificial neurons that receive input data, process information, identify patterns, and produce an output. Neural networks can learn from examples and improve their predictions during training.

The basic structure contains an **Input Layer**, one or more **Hidden Layers**, and an **Output Layer**. The input layer receives features, hidden layers process the information and learn patterns, and the output layer produces the final prediction or result. The network uses **weights, biases, and activation functions** while processing information.

Neural Networks are used in **image recognition, speech recognition, Natural Language Processing (NLP), medical diagnosis, recommendation systems, classification, and prediction**. Their ability to learn patterns from data makes them useful for many real-world problems.

BASIC STRUCTURE

A basic neural network consists of three main layers. Each layer performs a specific role in processing the input data and producing the final output.

- **Input Layer**

The input layer receives the data or features given to the neural network. For example, in an Iris flower classification system, sepal length, sepal width, petal length, and petal width can be given as input.

- **. Hidden Layer**

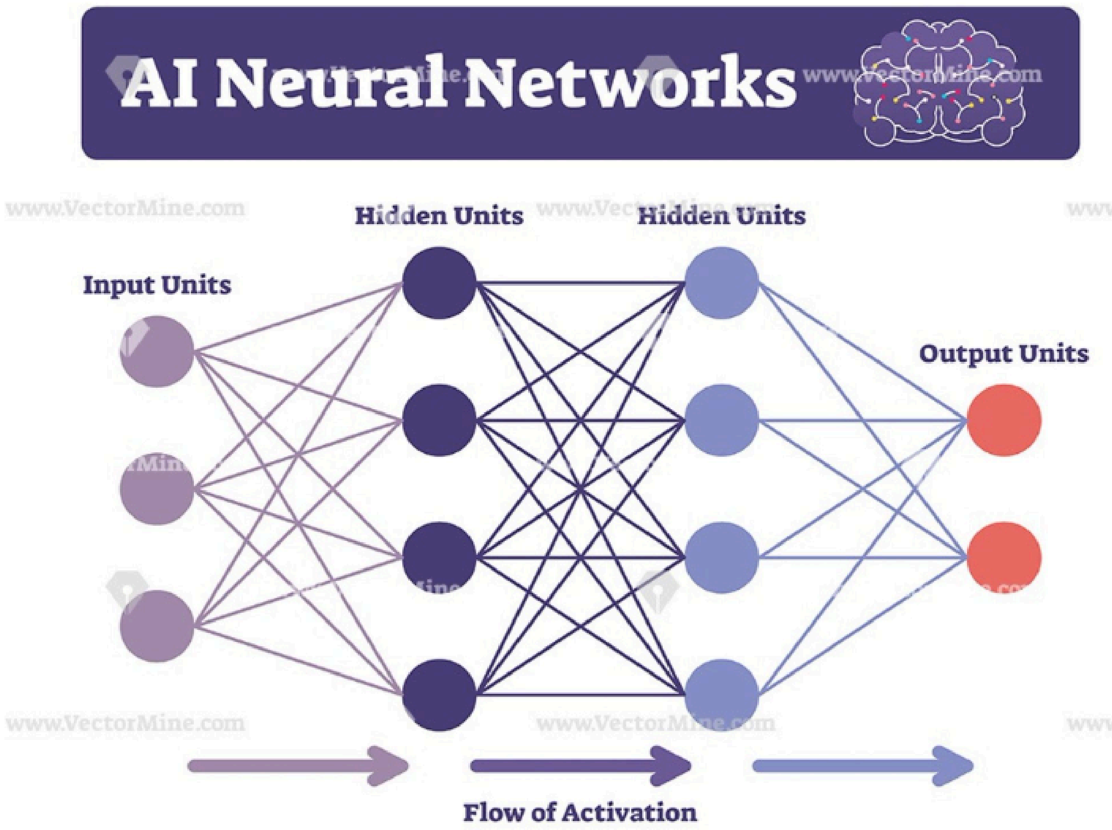
The hidden layer is located between the input layer and the output layer. It processes the input data and learns important patterns. A neural network may contain one or more hidden layers.

- **3. Output Layer**

The output layer produces the final prediction or result of the neural network. The output may be a classlabel, probability, or numerical value depending on the problem.

General structure: Input Layer → Hidden Layer(s) → Output Layer

Neural networks use mathematical operations, weights, biases, and activation functions to learn from data and make predictions.

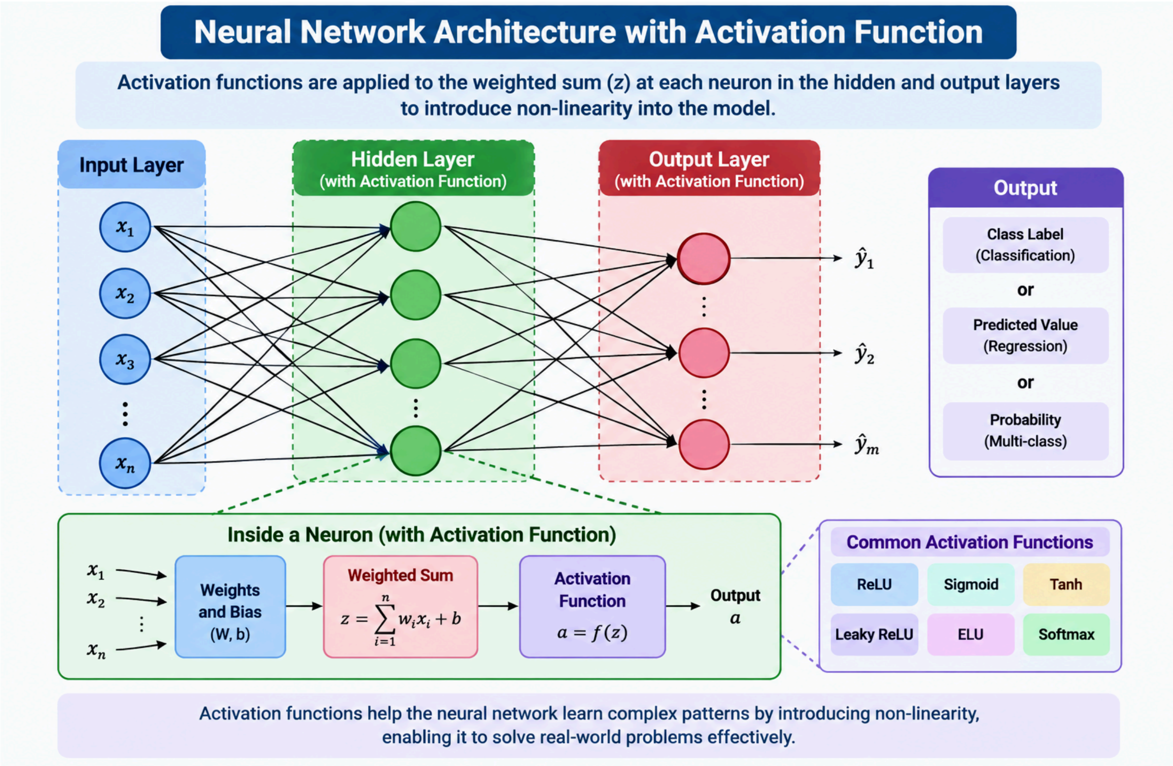


2. ACTIVATION FUNCTION

An **Activation Function** is a mathematical function used in a neural network to determine the output of a neuron. It is applied after calculating the weighted sum of inputs and bias. Activation functions introduce non-linearity, allowing the network to learn complex relationships.

Formula: $y = f(\Sigma wx + b)$
where x = input, w = weight, b = bias, f = activation function, and y = output.

3. TYPES OF ACTIVATION FUNCTIONS



❖ Binary Step Function

Produces 0 or 1 based on a threshold. Formula: f(x)=1 if x≥0, otherwise f(x)=0. It is useful for basic binary decisions. It is simple, but not suitable for complex neural networks.

Formula

f(x) = 1, if x ≥ 0

f(x) = 0, if x < 0

Advantages

- Simple and easy to understand
- Useful for basic binary decisions

Disadvantage

- Not suitable for complex neural networks

❖ Linear Activation Function

Produces an output directly proportional to the input. Formula: f(x)=x. It can be used for some regression problems, but it cannot learn complex nonlinear relationships and is not suitable for hidden layers of deep neural networks.

Formula

$f(x) = x$

The output changes directly according to the input value.

Advantages

- Simple mathematical calculation
- Useful in some regression problems

Disadvantages

- Cannot learn complex non-linear relationships
- Not suitable for hidden layers in deep neural networks

❖ Sigmoid Function

Converts an input into a value between 0 and 1. Formula: f(x)=1/(1+e^-x). It is commonly used for binary classification and probability prediction. It is smooth and differentiable, but can suffer from the vanishing-gradient problem.

Formula

$f(x) = 1 / (1 + e^{-x})$

Because its output is between 0 and 1, it is commonly used in **binary classification** problems. The output can also be interpreted as a probability.

Advantages

- Output is between 0 and 1
- Useful for binary classification
- Smooth and differentiable

Disadvantage

Can suffer from the **vanishing gradient problem**

❖ Tanh Function

Produces values between -1 and +1. Formula: (e^x−e^-x)/(e^x+e^-x). It is zero-centered and useful when positive and negative values are helpful. A limitation is the vanishing-gradient problem.

Formula

$f(x) = (e^x - e^{-x}) / (e^x + e^{-x})$

Tanh is similar to the Sigmoid function, but its output is **centered around zero**.

Advantages

- Output is centered around zero
- Useful for learning both positive and negative values

Disadvantage

- ❖ Can suffer from the **vanishing gradient problem**

❖ ReLU

ReLU means Rectified Linear Unit. Formula: f(x)=max(0,x). Positive inputs are passed unchanged and negative inputs become zero. It is widely used in deep learning because it is simple, fast, and computationally efficient. A limitation is the dying-ReLU problem.

Formula

$f(x) = \max(0, x)$

Advantages

- Simple and fast
- Computationally efficient
- Helps reduce the vanishing gradient problem

Disadvantage

- Can suffer from the **dying ReLU problem**

❖ Leaky ReLU

Leaky ReLU is an improved version of ReLU that allows a small negative output for negative inputs. Formula: f(x)=x if x>0, otherwise αx. It helps reduce the dying-ReLU problem and is useful in hidden layers of deep neural networks

. Formula

$f(x) = x$, if $x > 0$

$f(x) = \alpha x$, if $x \leq 0$

This allows negative inputs to contribute a small value and helps reduce the **dying ReLU problem**.

❖ Softmax

Softmax is mainly used for multiclass classification. It converts output values into probabilities whose total is 1. For example, Cat=0.70, Dog=0.20, Bird=0.10. The class with the highest probability is selected as the prediction

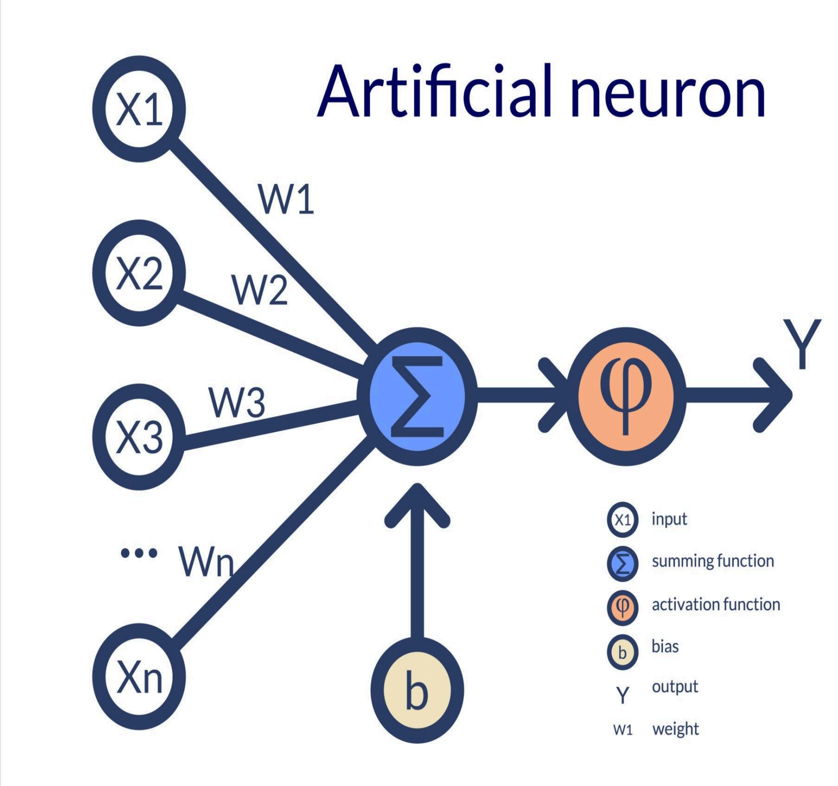
. For example, an image classification model may produce:

- Cat = 0.70
- Dog = 0.20
- Bird = 0.10

The class with the **highest probability** is selected as the final prediction. In this example, the predicted class is **Cat**.

4. TERMINOLOGY OF NEURAL NETWORKS

Understanding the basic terminology of Neural Networks is important for understanding how neural networks work.



1.Neuron

A **Neuron** is the basic processing unit of a Neural Network. It receives input values, performs mathematical calculations, applies an activation function, and produces an output.

2. Input

Input refers to the data given to the Neural Network for processing. Inputs are also called **features**.

3. Feature

A **Feature** is an individual characteristic or measurable property of input data that is used for prediction.

For example, in Iris flower classification, flower measurements can be used as input features.

4. Weight

A **Weight** represents the importance of an input. Each connection between neurons has an associated weight.

During training, the Neural Network adjusts the weights to improve its predictions.

5. Bias

A **Bias** is an additional value added to the weighted sum of inputs. Bias helps the Neural Network adjust its output and learn patterns more effectively.

6. Input Layer

The **Input Layer** is the first layer of a Neural Network. It receives the input data and passes it to the hidden layer.

7. Hidden Layer

A **Hidden Layer** is located between the input layer and output layer. It processes input data and learns complex patterns.

A Neural Network may contain one or more hidden layers. Deep Neural Networks contain multiple hidden layers.

8. Output Layer

The **Output Layer** is the final layer of a Neural Network. It produces the final prediction or result.

The output can be:

- A class label
- A probability
- A numerical value

9. Forward Propagation

Forward Propagation is the process of passing input data through the Neural Network from the input layer to the output layer.

The basic flow is:

Input → Hidden Layer → Output

During forward propagation, the network uses weights, biases, and activation functions to calculate the output.

10. Loss Function

A **Loss Function** measures the difference between the actual output and the predicted output.

The main objective of training a Neural Network is to **reduce or minimize the loss**.

11. Backpropagation

Backpropagation is the process used to update the weights and biases of a Neural Network based on the error produced.

The error is propagated backward through the network, and the weights are adjusted to improve future predictions.

12. Epoch

An **Epoch** represents one complete pass of the entire training dataset through the Neural Network.

For example:

10 Epochs = The complete dataset is processed 10 times.

13. Learning Rate

The **Learning Rate** determines how much the weights should be changed during the training process.

- Very high learning rate → Training may become unstable
- Very low learning rate → Training may become slow

14. Gradient Descent

Gradient Descent is an optimization algorithm used to minimize the loss function.

It helps the Neural Network find better values for **weights and biases** during training.

5. CONCLUSION

Neural Networks are an important part of Artificial Intelligence and Deep Learning. They learn patterns and relationships from data using interconnected neurons, weights, biases, layers, and activation functions.

Different activation functions such as Binary Step, Linear, Sigmoid, Tanh, ReLU, Leaky ReLU, and Softmax are used according to the problem. Understanding neural-network terminology such as neurons, inputs, weights, bias, layers, forward propagation, loss, backpropagation, epochs, learning rate, and gradient descent helps us understand how a neural network learns and makes predictions.